

Assignment: statistic vocabularies

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Statistic tables and first steps

At <https://zenodo.org/records/15681384> you will find two sets of statistic tables used in the experiments of the STAR paper [GEG⁺26]. One set of tables has more than 7 000 tables; the other is the subset of the first that contains the 2 000 smallest tables. A sample table is pictured below:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Time frequency	Unit of measure	Type of vessel	Transport coverage	Geopolitical entity (reporting)	Time												
2	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Austria	2012.0	2013.0	2014.0	2015.0	2016.0	2017.0	2018.0	2019.0	2020.0	2021.0	2022.0	2023.0	
3	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Belgium	2191.0	2393.0	2177.0	1806.0	1962.0	2022.0	1489.0	1715.0	1806.0	1506.0	1233.0	1187.0	
4	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Bulgaria	10420.0	10385.0	10481.0	10428.0	10201.0	11098.0	7988.0	7787.0	7388.0	8180.0	7959.0	6952.0	
5	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Czechia	5349.0	5374.0	5074.0	5095.0	5477.0	5279.0	4858.0	5901.0	6295.0	5792.0	3787.0	4253.0	
6	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Germany	38.0	25.0	27.0	33.0	36.0	25.0	23.0	32.0	38.0	22.0	20.0	18.0	
7	Annual	Million tonne-kilometre (TKM)	Total	Total transport	European Union - 27 countries (from 2020)	58488.0	60070.0	59893.0	55315.0	54347.0	58518.0	46991.0	50919.0	46338.0	48197.0	44117.0	41206.0	
8	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Finland	148832.0	151305.0	149431.0	145505.0	145287.0	145986.0	130051.0	138808.0	130478.0	135047.0	121971.0	116405.0	
9	Annual	Million tonne-kilometre (TKM)	Total	Total transport	France								122.0	127.0	148.0	184.0	221.0	
10	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Croatia	8981.0	9082.0	8770.0	8484.0	8310.0	7515.0	7255.0	7096.0	6998.0	7265.0	6993.0	5920.0	
11	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Hungary	772.0	771.0	716.0	879.0	836.0	813.0	678.0	835.0	903.0	841.0	595.0	486.0	
12	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Luxembourg	1982.0	1824.0	1811.0	1824.0	1975.0	1992.0	1608.0	2120.0	1998.0	1873.0	1508.0	1316.0	
13	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Netherlands	290.0	313.0	285.0	235.0	190.0	195.0	295.0	229.0	201.0	217.0	200.0	165.0	
14	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Poland	46681.0	47688.0	48252.0	46912.0	47657.0	47960.0	45886.0	46194.0	44106.0	46591.0	44551.0	41631.0	
15	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Romania	131.0	91.0	110.0	88.0	108.0	115.0	125.0	87.0	77.0	54.0	40.0	29.0	
16	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Sweden	12520.0	12242.0	11780.0	13168.0	13153.0	12517.0	12261.0	13957.0	13868.0	13522.0	10757.0	11947.0	
17	Annual	Million tonne-kilometre (TKM)	Total	Total transport	Slovakia											134.0	134.0	
18	Annual	Thousand tonnes	Total	Total transport	Austria	885.0	1006.0	955.0	741.0	903.0	933.0	778.0	937.0	834.0	839.0	680.0	641.0	
19	Annual	Thousand tonnes	Total	Total transport	Belgium	10714.0	10710.0	10122.0	8599.0	9071.0	9620.0	7202.0	8512.0	8247.0	8271.0	8374.0	8047.0	
20	Annual	Thousand tonnes	Total	Total transport	Bulgaria	190288.0	187404.0	190303.0	188158.0	192938.0	201129.0	151872.0	155695.0	156133.0	166160.0	183023.0	159990.0	
21	Annual	Thousand tonnes	Total	Total transport	Czechia	16378.0	16726.0	16822.0	17201.0	17487.0	16247.0	15482.0	18735.0	18924.0	18993.0	12546.0	13428.0	
22	Annual	Thousand tonnes	Total	Total transport	Germany	838.0	808.0	802.0	850.0	832.0	510.0	390.0	779.0	397.0	344.0	317.0	376.0	
23	Annual	Thousand tonnes	Total	Total transport	European Union - 27 countries (from 2020)	223170.0	226864.0	228489.0	221389.0	221349.0	222751.0	197904.0	205066.0	188022.0	195093.0	182448.0	171689.0	
24	Annual	Thousand tonnes	Total	Total transport	Finland	335042.0	333950.0	347177.0	340791.0	345754.0	351299.0	303784.0	314914.0	495590.0	513925.0	488144.0	468489.0	
25	Annual	Thousand tonnes	Total	Total transport	France	68542.0	68059.0	65139.0	62068.0	65189.0	63255.0	59529.0	64207.0	59979.0	57109.0	53201.0	47333.0	
26	Annual	Thousand tonnes	Total	Total transport	Croatia	5634.0	5623.0	5277.0	6442.0	6405.0	6221.0	5182.0	6491.0	7077.0	6548.0	4626.0	3785.0	
27	Annual	Thousand tonnes	Total	Total transport	Hungary	8135.0	7857.0	7825.0	8163.0	8224.0	8414.0	6826.0	8592.0	8803.0	8022.0	5921.0	5206.0	
28	Annual	Thousand tonnes	Total	Total transport	Luxembourg	8506.0	8987.0	8390.0	7106.0	6035.0	6163.0	5741.0	6433.0	5755.0	6221.0	5652.0	4745.0	
29	Annual	Thousand tonnes	Total	Total transport	Netherlands	343020.0	346426.0	357758.0	340717.0	354587.0	359356.0	349480.0	347774.0	335982.0	357694.0	345478.0	327479.0	
30	Annual	Thousand tonnes	Total	Total transport	Poland	2574.0	3185.0	5896.0	5036.0	3911.0	3604.0	3126.0	2870.0	2517.0	2075.0	918.0	587.0	
31	Annual	Thousand tonnes	Total	Total transport	Romania	27946.0	26858.0	27834.0	30020.0	30484.0	29045.0	29714.0	33261.0	30518.0	32120.0	28620.0	32102.0	
32	Annual	Thousand tonnes	Total	Total transport	Sweden											2450.0	1281.0	
33	Annual	Thousand tonnes	Total	Total transport	Slovakia	8242.0	8107.0	7010.0	5721.0	6758.0	6886.0	5567.0	6430.0	6004.0	6013.0	5174.0	4670.0	
34																		
35																		
36																		
37																		

The *leftmost columns* and *top rows* contain dimension names, dimension values, measure names, and/or units in which the measure is taken. The rest of each table contains measure values (numbers). Sometimes, next to each number appears a single-letter indicator, such as **i**, or **b**, etc. These indicators denote, e.g., if a number is temporary, incomplete, etc. They can be ignored.

At <https://drive.google.com/file/d/1gp0P1QT2huUe6TuirorjGMpbtVq8-Eki/view?usp=sharing> you will find the natural-language title of every table.

- On the top line of each table, at the right of the text (non-number) cells, these (Eurostat) tables almost systematically feature *dates*; most often, these are years, but other time units may also appear (at the granularity of day, month, quarter, etc.) Build $D(t)$, be the set of time intervals appearing on the top line of each table t . Rough heuristics or pattern matching are OK, e.g., consider that any number appearing there, which is between 1900 and 2100, is a year, etc.
- From each table t , extract $S(t)$, the set of all strings (not numbers) that appear in its leftmost columns. $S(t)$ is disjoint from $D(t)$; also, $S(t)$ should be duplicate-free.
- Identify the geographical units in each table's vocabulary: $Geo(t) \subset S(t)$. These are units whose names appear in the NUTS geographic dictionary <https://ec.europa.eu/eurostat/web/nuts> used by Eurostat. Let $V(t) = S(t) \setminus Geo(t)$.
- Let $\tau(t)$ be the title of table t . Process $\tau(t)$ like we did above for $S(t)$, i.e., separate from it the dates and the geographic locations. Add the remainder to $V(t)$, for each table t .
- Build $\mathcal{V} = \bigcup_t V(t)$ be the total vocabulary (set of distinct terms) appearing in one or more tables.

Measures, dimension names, dimension values, and unit information

6. Terms in $\mathcal{V}$ fall into one of these categories:

- Measures, e.g., "Economic accounts for agriculture - values at current prices"
- Dimension names, e.g., "Type of vessel"
- Dimension values, e.g., "Fishing vessel"
- Unit information, e.g., "Annual", "Million tonne-kilometre (TKM)", etc.

Propose and implement a method to partition $\mathcal{V}$ into four parts: **measures** $\mathcal{M}$, **dimension names** $\mathcal{N}$, **dimension values** $\mathcal{A}$, and **unit information** $\mathcal{U}$.

- You can work on $\mathcal{V}$ as a whole, or on each $V(t)$ individually (however, consider that there may be significant overlap across $V(t)$'s; try not to repeatedly do the same work).
- If needed, you can add more categories to the four above, or just add an "Other" category $\mathcal{O}$. If you do this, explain the need.

Structuring the measures

This step will be based on the set $\mathcal{M}$ identified in the previous section. The goal is to **structure** the measures, i.e., organize them and if possible identify meaningful relationships against them.

7. Propose an approach for grouping (clustering) the measures in $\mathcal{M}$ in domains, e.g., "agriculture", "transport", "industry" etc. You can do this with the help of an explicit list of domains built by hand or retrieved from an existing source, or by emerging the categories from $\mathcal{M}$ itself.
8. (Bonus) Propose ways to identify semantic relationships between pairs of individual measures.

Throughout the assignment, you can use any method (string embeddings, dictionaries, zero- or few-shot prompting of LLMs, etc.) If LLMs are used, ensure that code is present to prevent "hallucinatings" (introduction of terms that are not in $\mathcal{V}$, and are not direct or trivial normalizations of $\mathcal{V}$ terms, either).

What to turn in

The code together with complete installation instructions should be in a **publicly accessible Git repository**.

Document the usage of any resource (static, or LLM, etc.) necessary. Include the prompts used in separate files.

The repository should also contain your proposed $\mathcal{M}$, $\mathcal{N}$, $\mathcal{A}$ and $\mathcal{U}$ in **four files**: `measures.csv`, `dimension_names.csv`, `dimension_values.csv` and `units.csv`.

The repository should also contain a **report** following the numbering of the assignment, pointing to the respective code and explaining the method used.

You should **propose, implement, and describe (in the report) a quality evaluation** for steps 6, 7 and 8.

References

- [GEG⁺26] Antoine Gauquier, Simon Ebel, Helena Galhardas, Théo Galizzi, Ioana Manolescu, Aurélien Peden, and Pierre Senellart. Efficient and Scalable Search for Statistics. In *ICDE 2026 - 42nd IEEE International Conference on Data Engineering*, Montréal, Canada, May 2026.